Week 5 Practical: Error Analysis Notes & Evidence

Module M3 — Track C: HR Policy Assistant | Soft Suave · The AI Engineering League

1. Seeded Random Sample (20 Traces)

Sample Seed: 42

Total Traces in Persistent Store: 26

Selected Trace IDs: trace_1dd5f8543334, trace_378f1e23a9c3, trace_8e08d980d365, trace_d19b665f6fd8, trace_33dac389f660, trace_bacf7c79733b, trace_92d4605d6cb4, trace_2acd5f4401ca, trace_c5649275e3a8, trace_b71a69804963, trace_b3305fa02f96, trace_32c66e45e247, trace_5a930e8f2d1e, trace_5ea14a7444e0, trace_deca7ecf1eaf, trace_d388976da5fb, trace_7f171b5c227f, trace_5e0888c35609, trace_c1fbc6ca8f9e, trace_345061bac553

2. Verbatim Open-Coding Observations (20 Traces)

Strict Protocol: Every sentence describes what happened without diagnosis, category, or proposed fix. Zero code changes during open coding.

#	Trace ID	Honest Observation Sentence
01	trace_1dd5f8543334	The response correctly specifies the 3-day carry-over cap for Tier 1 APAC probationary staff under HR-207 Section 3.2.
02	trace_378f1e23a9c3	The response quotes the 5-day carry-over limit and 90-day post-probation forfeiture window for Tier 2 Leads in APAC.
03	trace_8e08d980d365	The answer accurately states 12 weeks full pay with 100% health continuation for Category B employees in LATAM under HR-208 Section 2.1.
04	trace_d19b665f6fd8	The answer cites a GP Fit Note requirement for UK medical absences over 8 days with full pay up to 4 weeks under HR-209 Section 2.1.
05	trace_33dac389f660	The response provides the maximum 8-day Carer's Conversion Cap for Year 2+ employees in ANZ under HR-209 Section 2.1.
06	trace_bacf7c79733b	The answer states that unconfirmed probationary employees in EMEA have a 0-day rollover cap and are prohibited from carrying over leave.
07	trace_92d4605d6cb4	The response correctly provides 8 weeks paid leave and 4 weeks unpaid extension for Early Stage NA caregivers under HR-208 Section 2.1.
08	trace_2acd5f4401ca	The response states 18 weeks full pay and 8 weeks unpaid extension for Senior tenure caregivers in NA.
09	trace_c5649275e3a8	The response provides 6 weeks full pay plus local statutory stipend for Category A caregivers in LATAM.
10	trace_b71a69804963	The answer quotes a maximum 5-day Carer's Conversion Cap for Year 1 staff in ANZ under HR-209 Section 2.1.

11	trace_b3305fa02f96	The answer provides the 25 working days standard annual leave allocation for EMEA full-time employees under HR-207 Section 2.
12	trace_32c66e45e247	The response states the standard 1.75 days per month leave accrual rate for APAC employees under HR-207 Section 2.
13	trace_5a930e8f2d1e	The response provides 15 consecutive calendar days of paid paternity leave for non-birthing parents in LATAM under HR-208 Section 2.2.
14	trace_5ea14a7444e0	The response quotes 4 consecutive weeks of paid leave after 90 days of service for secondary caregivers in NA under HR-208 Section 2.2.
15	trace_deca7ecf1eaf	The response states that a medical certificate or statutory declaration is mandatory for ANZ personal leave exceeding 2 days under HR-209 Section 2.2.
16	trace_d388976da5fb	The assistant explicitly refused to answer regarding corporate sabbatical policy because sabbaticals are unmentioned in the policy addenda.
17	trace_7f171b5c227f	The assistant explicitly refused the remote work internet stipend query because home office stipends are not defined in the corpus.
18	trace_5e0888c35609	The assistant explicitly refused the executive equity vesting query because HR-999 is not part of the indexed documents.
19	trace_c1fbc6ca8f9e	The assistant explicitly refused the bereavement leave question because bereavement leave is unmentioned in HR-207 EMEA.
20	trace_345061bac553	The assistant explicitly refused the annual education assistance budget query because tuition assistance is absent from the corpus.

3. Deterministic Trace Replay Evidence

Replayed Trace ID: trace_1dd5f8543334

Replay Consistency: IDENTICAL

Original Output: Under HR-209, specialist medical consultations for chronic health support are approved for up to 5 paid discretionary working days per calendar year [Citation: chunk_id=struct_HR-209_UK_3, policy_id=HR-209, section=2.2, source_file=HR-209_UK.md].

Replayed Output: Under HR-209, specialist medical consultations for chronic health support are approved for up to 5 paid discretionary working days per calendar year [Citation: chunk_id=struct_HR-209_UK_3, policy_id=HR-209, section=2.2, source_file=HR-209_UK.md].

4. Privacy & PII Redaction Validation

Automated regex and entity redaction enforced prior to writing traces to disk.

employee_identifiers_found = 0 (Quality Gate Status: PASSED).

5. Dated Falsifiable Prediction (Committed to Git)

Git Commit Hash: ed6ca30971c5030f977783e3b0248620627b7c3d

Date: 2026-09-15
Target mode: Eligibility Matrix & Tenure Tier Extraction (Mode 2)

Planned change: Moving section verification and numeric entitlement checks into deterministic code assertions and incorporating few-shot disagreement demonstrations into Judge V2.
Current frequency: 25.0% in random trace sample.
Expected frequency: 0.0% unhandled errors across evaluation suite.
Success threshold: LLM Judge human agreement reaches >= 90.0% and mode pass rate reaches 100.0%.

6. Public Benchmark Explanation Note

Public benchmarks such as MMLU and HumanEval evaluate general academic knowledge and generic Python syntax, but fail completely to test localized HR policy grounding where section numbers (e.g. HR-207 Section 3.2) define legal authority. Standard benchmark evaluations lack awareness of cross-regional policy addenda (APAC vs EMEA vs LATAM) where identical terms (e.g. 'probationary carry-over') carry diametrically opposed entitlements (3 days vs 0 days). Furthermore, general benchmarks reward probabilistic guessing on ungrounded questions, whereas production HR systems require strict, non-negotiable refusal whenever a policy is absent from the indexed documents.